

The 168 Quantum of Non-Associativity in the Cayley–Dickson Tower

Demetrios C. Agourakis

Biomaterials and Regenerative Medicine Post-Graduate Program,
Pontifícia Universidade Católica de São Paulo (PUC-SP), Sorocaba, SP, Brazil
ORCID: 0009-0001-8671-8878

March 2026

Correspondence: demetrios@agourakis.med.br

Abstract. We study the number T_k of ordered triples of imaginary basis elements with nonzero associator in the 2^k -dimensional Cayley–Dickson algebra. It is classical that $T_3 = 168 = |\text{PSL}(2, 7)|$ for the octonions. By exhaustive computation we establish T_k for $k = 3, \dots, 7$ (through dimension 128), and conjecture $T_k = 168(P_k - 4P_{k-1})$ where $P_k = (2^k - 1)(2^{k-1} - 1)(2^{k-2} - 1)/21$ counts Fano subplanes in $\text{PG}(k - 1, 2)$. This formula is verified at five levels. We prove that the associator norm dichotomy $\|[e_i, e_j, e_k]\| \in \{0, 2\}$ holds for *all* Cayley–Dickson algebras, as a consequence of the bivalence of the Cayley–Dickson sign function and the associativity of bitwise exclusive-or. We derive that $T_k/(2^k - 1)^3 \rightarrow 1/2$ asymptotically: half of all basis triples are non-associative. The number of Fano subplanes grows by a factor of 7 per doubling, yielding a combinatorial scaling exponent $\log_2 7 \approx 2.807$.

Keywords: Cayley–Dickson algebras, non-associativity, Fano plane, projective geometry, $\text{PSL}(2, 7)$

MSC 2020: 17A75, 20B25, 17D05, 51E20

1. Introduction

The Cayley–Dickson construction produces a tower of real algebras by iterated doubling:

$$\mathbb{R} \rightarrow \mathbb{C} \rightarrow \mathbb{H} \rightarrow \mathbb{O} \rightarrow \mathbb{S} \rightarrow \mathbb{T} \rightarrow \dots \quad (1)$$

At each step, a structural property is lost: commutativity at the quaternions $\mathbb{H}$, associativity at the octonions $\mathbb{O}$, and the division property at the sedenions $\mathbb{S}$ [1, 2]. In a companion paper [3], we proved that the number of ordered triples $(i, j, k) \in \{1, \dots, 7\}^3$ of imaginary octonion basis elements with nonzero associator is exactly $T_3 = 168 = |\text{PSL}(2, 7)|$, following from the regular action of $\text{Aut}(\text{Fano})$ on ordered non-collinear triples.

A natural question arises: *does the 168-divisibility persist in higher Cayley–Dickson algebras?* In the sedenions ($2^4 = 16$ dimensions, $15^3 = 3375$ basis triples) and beyond, new phenomena appear — zero divisors, non-alternativity, and a proliferation of Fano subplanes in the associated projective geometry $\text{PG}(k - 1, 2)$. The nonzero associator count T_k grows rapidly, but its relationship to 168 has not been studied.

In this paper we address this gap. By exhaustive computation through $k = 6$ (dimension 64, requiring classification of $63^3 = 250\,047$ triples), we establish that T_k is a multiple of 168 at every level tested, and conjecture a closed formula relating T_k to the number of Fano subplanes in $\text{PG}(k - 1, 2)$. We derive the asymptotic fraction of non-associative triples and identify a combinatorial scaling exponent for the Fano subplane structure.

Contributions.

1. T_k computed exhaustively for $k = 3, 4, 5, 6, 7$ (Theorem 1).
2. A conjectured closed formula $T_k = 168(P_k - 4P_{k-1})$ verified at five levels (Conjecture 2).
3. Asymptotic analysis: $T_k/N^3 \rightarrow 1/2$ (Proposition 3).
4. The binary norm property $\| [e_i, e_j, e_k] \| \in \{0, 2\}$ *proven* for all k (Theorem 4).
5. Fano subplane branching exponent $\log_2 7 \approx 2.807$ (Proposition 5).
6. A sign function method for computing T_k without high-dimensional array operations (§4).

2. Preliminaries

2.1. Cayley–Dickson algebras

Given a real algebra A with conjugation $\bar{\cdot}$, the Cayley–Dickson doubling defines a new algebra $A' = A \times A$ with multiplication:

$$(a, b)(c, d) = (ac - \bar{d}b, da + b\bar{c}) \quad (2)$$

Starting from $\mathbb{R}$, this yields $\mathbb{C}$, $\mathbb{H}$, $\mathbb{O}$, $\mathbb{S}$, and the 2^k -dimensional algebra A_k for each $k \geq 0$. The algebra A_k has $2^k - 1$ imaginary basis elements $e_1, \dots, e_{2^k-1}$, with $e_0 = 1$ the identity. The associator of three elements is $[a, b, c] = (ab)c - a(bc)$.

2.2. Fano subplanes in $\text{PG}(k-1, 2)$

The imaginary basis elements of A_k are naturally indexed by nonzero vectors of $\mathbb{F}_2^k$, and the projective space $\text{PG}(k-1, 2)$ has $2^k - 1$ points. A **Fano subplane** is a subgeometry isomorphic to $\text{PG}(2, 2)$ — the Fano plane, with 7 points and 7 lines. The number of Fano subplanes in $\text{PG}(k-1, 2)$ is:

$$P_k = \frac{(2^k - 1)(2^{k-1} - 1)(2^{k-2} - 1)}{21} \quad (3)$$

with $P_2 = 0$, $P_3 = 1$, $P_4 = 15$, $P_5 = 155$, $P_6 = 1395$ [4].

3. Results

3.1. Exhaustive computation of T_k

We define T_k as the number of ordered triples (i, j, k) with $i, j, k \in \{1, \dots, 2^k - 1\}$ such that $[e_i, e_j, e_k] \neq 0$ in A_k .

Theorem 1. *The nonzero basis associator counts for $k = 3, 4, 5, 6, 7$ are:*

k	Dim	$N = 2^k - 1$	N^3	T_k	$T_k/168$
3	8	7	343	168	1
4	16	15	3 375	1 848	11
5	32	31	29 791	15 960	95
6	64	63	250 047	130 200	775
7	128	127	2 048 383	1 046 808	6 231

Each value is verified by exhaustive classification of all N^3 ordered triples.

For $k = 3$, the result is classical ($168 =$ number of ordered non-collinear triples in the Fano plane [1]). For $k = 4$, the value $T_4 = 1848 = 11 \times 168$ was reported in [3]. The values T_5 , T_6 , and T_7 are new.

For $k \leq 6$, each associator is computed via four Cayley–Dickson multiplications (recursively through k levels of doubling); two independent runs produced identical results for $k = 6$. For $k = 7$, we use the sign function method (§3.3): each triple is classified by four evaluations of σ (integer recursion), avoiding high-dimensional array operations entirely.

3.2. A conjectured closed formula

Conjecture 2. For all $k \geq 3$:

$$T_k = 168(P_k - 4P_{k-1}) \quad (4)$$

where P_k is given by Equation 3.

Verification. The conjecture is confirmed at all four computed levels:

k	P_k	P_{k-1}	$168(P_k - 4P_{k-1})$	Match
3	1	0	$168 \times 1 = 168$	✓
4	15	1	$168 \times 11 = 1848$	✓
5	155	15	$168 \times 95 = 15\,960$	✓
6	1395	155	$168 \times 775 = 130\,200$	✓
7	11\,811	1395	$168 \times 6231 = 1\,046\,808$	✓

The factor $P_k - 4P_{k-1}$ admits a structural interpretation. Each Fano subplane in $\text{PG}(k-1, 2)$ supports a copy of the octonion multiplication table. In the Cayley–Dickson algebra A_k , these subplanes split into two classes: *octonionic* subplanes (where the restricted algebra is alternative, contributing 168 nonzero associator triples each) and *quasi-octonionic* subplanes (where alternativity fails, contributing fewer). The structure of this decomposition has been studied for sedenions by Cawagas [5]. A proof of Equation 4 for general k would require establishing that quasi-octonionic subplanes contribute exactly 72 nonzero triples each and that their count is $7P_{k-1}$. We leave this as an open problem.

3.3. Asymptotic analysis

Proposition 3. If Conjecture 2 holds, then $T_k/(2^k - 1)^3 \rightarrow 1/2$ as $k \rightarrow \infty$.

Proof. For large k , $P_k \approx 2^{3k-3}/21$ (from Equation 3). Then:

$$T_k \approx 168 \cdot (2^{3k-3}/21 - 4 \cdot 2^{3(k-1)-3}/21) = 168 \cdot 2^{3k-6} \cdot (8 - 4)/21 = 168 \cdot 4 \cdot 2^{3k-6}/21 = 2^{3k-1} \quad (5)$$

Since $(2^k - 1)^3 \approx 2^{3k}$, the ratio $T_k/(2^k - 1)^3 \rightarrow 2^{3k-1}/2^{3k} = 1/2$. $\square$

This means that asymptotically, *exactly half* of all ordered basis triples are non-associative. The convergence is visible from the data: the fraction is 0.490 at $k = 3$, 0.548 at $k = 4$, 0.536 at $k = 5$, 0.521 at $k = 6$, and 0.511 at $k = 7$, approaching 0.500.

3.4. Norm dichotomy

Theorem 4. For any $k \geq 1$ and any nonzero $i, j, m \in \mathbb{Z}_2^k$: $\| [e_i, e_j, e_m] \| \in \{0, 2\}$.

Proof. The Cayley–Dickson construction determines a sign function $\sigma : (\mathbb{Z}_2^k \setminus \{0\})^2 \rightarrow \{+1, -1\}$ such that $e_i \cdot e_j = \sigma(i, j) \cdot e_{i \oplus j}$. Expanding the associator:

$$(e_i e_j) e_m = \sigma(i, j) \cdot \sigma(i \oplus j, m) \cdot e_{(i \oplus j) \oplus m} \quad (6)$$

$$e_i (e_j e_m) = \sigma(j, m) \cdot \sigma(j, i \oplus m) \cdot e_{i \oplus (j \oplus m)} \quad (7)$$

Since $\oplus$ is associative on $\mathbb{Z}_2^k$, both parenthesisations yield the same basis element $e_{i \oplus j \oplus m}$. Let $\alpha = \sigma(i, j) \cdot \sigma(i \oplus j, m)$ and $\beta = \sigma(j, m) \cdot \sigma(j, i \oplus m)$. Since σ is bivalent, $\alpha, \beta \in \{+1, -1\}$, so $\alpha - \beta \in \{-2, 0, +2\}$. Therefore $[e_i, e_j, e_m] = (\alpha - \beta) \cdot e_{i \oplus j \oplus m}$ and $\| [e_i, e_j, e_m] \| \in \{0, 2\}$. $\square$

Remark. The proof is dimension-independent: it uses only the bivalence of σ and the associativity of $\oplus$. No properties specific to the octonions (alternativity, norm multiplicativity) are required. The result applies to all Cayley–Dickson algebras, including those with zero divisors.

3.5. Fano subplane branching exponent

Proposition 5. If Conjecture 2 holds and the quasi-octonionic plane count is $Q_k = 7P_{k-1}$, then the total number of Fano subplanes satisfies $P_k/P_{k-1} \rightarrow 2^3 = 8$ as $k \rightarrow \infty$, and the number of non-associative

triples per Fano subplane scales with branching factor 7, yielding a combinatorial scaling exponent of $\log_2 7 \approx 2.807$.

Proof. From $P_k \approx 2^{3k-3}/21$, we have $P_k/P_{k-1} \approx 8$. Each doubling creates $P_k - P_{k-1} \approx 7P_{k-1}$ new quasi-octonionic planes (and the P_{k-1} existing planes persist as octonionic). The branching factor for the non-octonionic structure is therefore 7, and the scaling exponent in the logarithmic base of the dimension (2^k) is $\log_2 7 \approx 2.807$. $\square$

The codimension of the Fano subplane structure in the three-dimensional triple space is $3 - \log_2 7 \approx 0.193$. This means the non-associative structure, while filling asymptotically half the triple space by measure, is organized around a Fano subplane skeleton of sub-integer dimension.

4. Computational methods

Two methods are used. For $k \leq 6$, associators are computed via full Cayley–Dickson multiplication: $(a, b)(c, d) = (ac - \bar{d}b, da + b\bar{c})$, recursing through k levels. For $k = 7$ (and potentially higher), we use a *sign function method* that avoids high-dimensional arrays entirely.

4.1. Sign function method

By Theorem 4, determining whether $[e_i, e_j, e_m] \neq 0$ requires only checking whether $\alpha \neq \beta$, where $\alpha = \sigma(i, j) \cdot \sigma(i \oplus j, m)$ and $\beta = \sigma(j, m) \cdot \sigma(i, j \oplus m)$. The sign function σ is computed by integer recursion through k levels (one comparison and one XOR per level), with no floating-point arithmetic and no array allocation. This reduces the per-triple cost from $O(2^k)$ (full multiplication) to $O(k)$ (sign recursion), enabling computation at dimensions where full multiplication is impractical.

At $k = 7$ ($127^3 = 2048383$ triples), the sign function method classifies all triples using only integer operations. The method scales to arbitrary k limited only by the $O(N^3) = O(2^{3k})$ enumeration cost.

Source code (including the recursive sign function implementation) is available at github.com/agourakis82/sounio, implemented in the Sounio programming language [6].

5. Open questions

1. **Prove Conjecture 2 for all $k \geq 3$.** The key step is establishing the octonionic/quasi-octonionic subplane decomposition: show that each Fano subplane in A_k is either octonionic (contributing 168 nonzero associator triples) or quasi-octonionic (contributing 72), and that the quasi-octonionic count is $7P_{k-1}$.
2. **Characterize the structure of the sign function.** Theorem 4 establishes that the norm dichotomy follows from the bivalence of σ . Can the nonzero count T_k be derived analytically from the properties of σ , without exhaustive enumeration?
3. **Characterize the sub-class distribution.** For sedenions, the nonzero associator triples decompose by index range (oct-oct-oct, sed-sed-sed, cross) with each sub-count a multiple of 168 [3]. Does this hold for $k \geq 5$?

6. Acknowledgments

The octonion base result ($T_3 = 168$) was established in joint work with M. Geranutti [3].

Bibliography

1. Baez, J. C.: The Octonions. *Bulletin of the American Mathematical Society*. 39, 145–205 (2002)
2. Moreno, G.: The zero divisors of the Cayley–Dickson algebras over the real numbers. *Boletín de la Sociedad Matemática Mexicana*. 4, 13–28 (1998)
3. Agourakis, D. C., Geranutti, M.: The 168 Count for Octonion Basis Associators and a Computational Cayley–Dickson Tower Extension. *Advances in Applied Clifford Algebras*. (2026)
4. Hirschfeld, J. W. P.: *Projective Geometries over Finite Fields*. Oxford University Press (1998)

5. Cawagas, R. E.: On the structure and zero divisors of the Cayley–Dickson sedenion algebra. *Discussiones Mathematicae — General Algebra and Applications*. 24, 251–265 (2004)
6. Agourakis, D. C.: Sounio: A systems programming language for epistemic computing, <https://github.com/agourakis82/sounio>